

Wiring, Cables and the Panels With Names

What the conductors are, how they are identified, and the installations that are reportable on sight.

Branch circuit wiring	The wiring running from the main panel through the walls to the outlets where current is used.
Receptacles	The devices connected to branch circuit wires that supply current to equipment. An outlet is the point in the wiring; the receptacle is the device fitted there.
Multiwire branch circuit	Three wires: two hot conductors and one shared neutral. The hots must sit on opposite phases or the shared neutral carries the sum of both loads and overheats.
Polarized receptacle	The hot side is the narrow slot. The wide slot is neutral.
Color code	Black or red is hot. White is neutral. Green or bare is equipment ground.

Aluminum

Aluminum expands, contracts and oxidises at its terminations, which loosens connections and makes heat. It is used primarily for high-draw single-appliance circuits and service entrances — electric ovens and ranges rather than general lighting and receptacles. It is marked AL on the conductor, and an antioxidant must be applied to all AL conductors.

YOU HAVE MISSED THIS 2 TIMES

Aluminum wiring is used primarily for which of the following?

Electric ovens.

The most likely aluminum installation to overheat is a receptacle serving an electric heater: a continuous heavy load through a connection type that loosens as it heats and cools.

Cable types worth naming

Knob and tube	Early ungrounded wiring. Its receptacles are not grounded, the insulation goes brittle with age, and it must never be covered by insulation because it sheds heat into open air.
BX	Cable in a metallic enclosure.
UF	The cable type listed for direct burial.
IC-rated fixture	A recessed light fixture that may be in contact with insulation. One that is not IC-rated may not.

Two brands you report by name

Federal Pacific Electric (FPE) Stab-Lok breakers have a documented history of failing to trip on overcurrent. Zinsco and Sylvania breakers can fuse to the bus and fail to trip. Both are reported as a safety concern with a recommendation for evaluation by a qualified electrician — you report and recommend evaluation, you do not declare that the panel must be replaced.

Double tap

Two conductors under one breaker terminal that is listed for only one. It is one of the most common panel findings there is, and it is a defect regardless of how tidy the work looks.

Say it back

Cover the page. Answer each one out loud.

1. What is branch circuit wiring, and how does an outlet differ from a receptacle?
2. What makes a multiwire branch circuit dangerous if it is wired wrong?
3. Which slot on a polarized receptacle is hot?
4. What is aluminum wiring primarily used for, and what must be applied to it?
5. Name the aluminum installation most likely to overheat.
6. Two facts about knob and tube: the receptacles, and the insulation.
7. UF, BX, IC — what does each one mean?
8. Name the two problem panel brands and exactly what you write.